Continuous Random Variables

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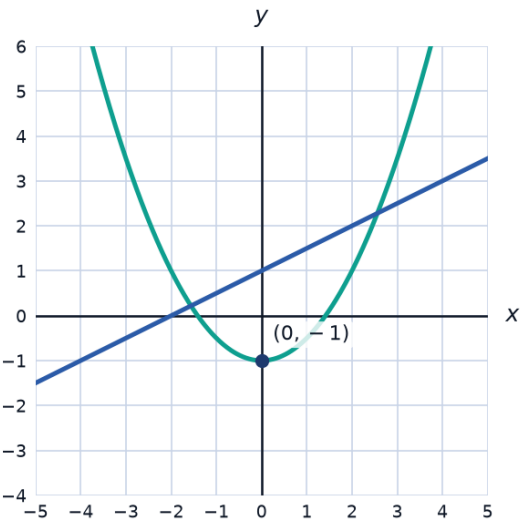
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Verifying a valid probability density function**

A continuous random variable X has probability density function $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. Verify that this is a valid pdf.

2 **[Foundational] • Target: Verifying a valid probability density function**

Determine the value of c such that $f(x) = c$ for $2 \le x \le 6$, and $f(x) = 0$ otherwise, is a valid pdf.



3 **[Foundational] • Target: Finding an unknown constant in a pdf**

A continuous random variable X has pdf $f(x) = kx$ for $0 \le x \le 5$, and $f(x) = 0$ otherwise. Find k.

4 **[Foundational] • Target: Finding an unknown constant in a pdf**

A continuous random variable X has pdf $f(x) = k(4 - x)$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. Find k.

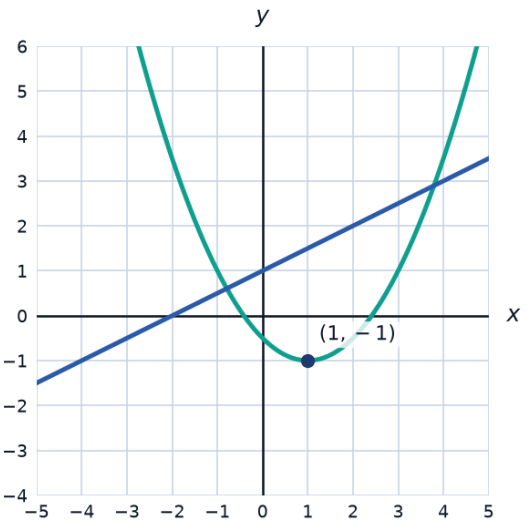
5 **[Foundational] • Target: Finding an unknown constant in a pdf**

A continuous random variable X has pdf $f(x) = kx^2$ for $0 \le x \le 3$, and $f(x) = 0$ otherwise. Find k.

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6 [Foundational] • Target: Finding probabilities by integrating the pdf

Using $f(x) = \frac{2}{25}x$ for $0 \le x \le 5$, find $P(X \le 2)$.



7 [Foundational] • Target: Finding probabilities by integrating the pdf

Using the same pdf, find $P(2 \le X \le 4)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding probabilities by integrating the pdf

Using $f(x) = \frac{1}{8}(4 - x)$ for $0 \le x \le 4$, find $P(X \ge 2)$.

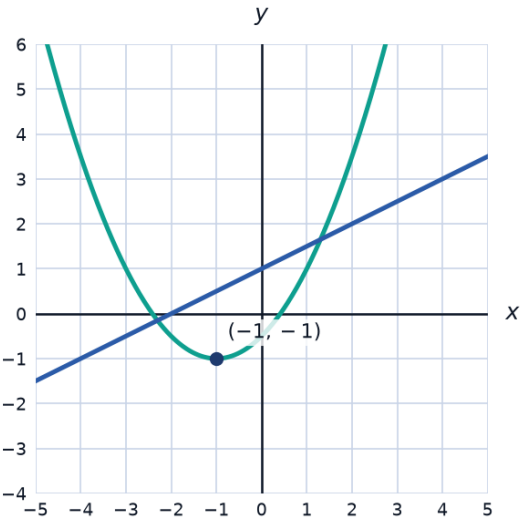
9 [Intermediate] • Target: Finding probabilities by integrating the pdf

Using $f(x) = \frac{1}{9}x^2$ for $0 \le x \le 3$, find $P(X \le 1)$.

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10 [Intermediate] • Target: The cumulative distribution function

For the pdf $f(x) = \frac{2}{25}x$ on $0 \le x \le 5$, find the cumulative distribution function $F(x)$.



11 [Intermediate] • Target: The cumulative distribution function

Using $F(x) = \frac{x^2}{25}$, find $P(X \le 3)$ and hence $P(X > 3)$.

12 [Intermediate] • Target: The cumulative distribution function

For the pdf $f(x) = \frac{1}{8}(4 - x)$ on $0 \le x \le 4$, find the cumulative distribution function $F(x)$.

13 [Intermediate] • Target: The cumulative distribution function

For the pdf $f(x) = \frac{1}{9}x^2$ on $0 \le x \le 3$, find the cumulative distribution function $F(x)$.

14 [Intermediate] • Target: Using the cdf to find probabilities

Given $F(x) = \frac{x^2}{25}$ for $0 \le x \le 5$, find $P(1 \le X \le 4)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

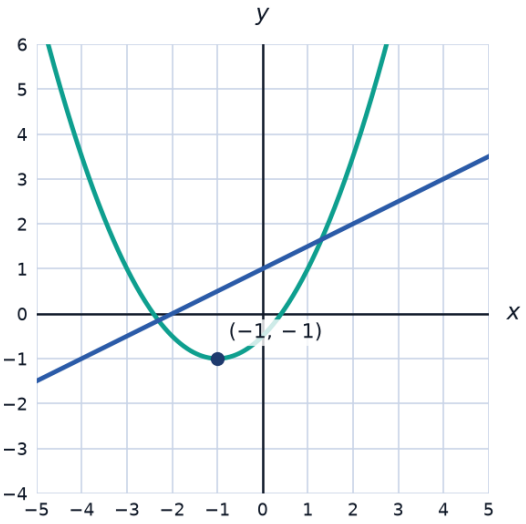
15 [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 14.01 Sampling & Estimation

Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$, find $P(1 \le X \le 3)$.

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16 [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 15.01 Hypothesis Testing

Using $F(x) = \frac{x^3}{27}$ for $0 \le x \le 3$, find $P(1 \le X \le 2)$.



17 [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 2.01 Exponentials & Logarithms

Using $f(x) = \frac{2}{25}x$ for $0 \le x \le 5$, find $E(X)$, using $E(X) = \int x f(x)\,dx$.

18 [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 3.01 Advanced Trigonometry

Using the uniform pdf $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, find $E(X)$, and verify it matches the midpoint of the interval.

19 [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 4.01 Differentiation

Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, find $E(X)$.

20 [Challenging] • Target: Variance of a continuous random variable • Cross-Lesson: 5.01 Integration

Using $f(x) = \frac{2}{25}x$ on $[0, 5]$ and $E(X) = \frac{10}{3}$ (found earlier), find $\mathrm{Var}(X)$, using $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Probability Density Functions (PDF)

A continuous random variable X has PDF $f(x) = kx(2 - x)$ for $0 \le x \le 2$, and 0 otherwise. Find the value of k.

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{8}$

d. 1

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22 [Foundational] • Target: Probability Density Functions (PDF)

For the random variable in the previous question ($k = 3/4$), calculate $P(0 \le X \le 1)$.

a. $\frac{1}{2}$

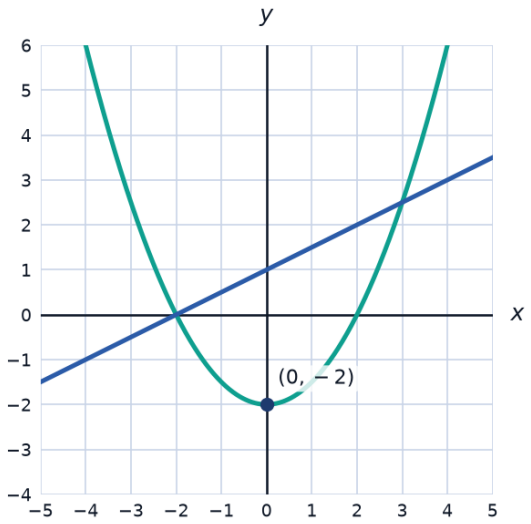
b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{5}{8}$

23 [Foundational] • Target: Probability Density Functions (PDF)

What is the probability that a continuous random variable takes any single exact value, such as $P(X = c)$?



a. 0

b. 1

c. $f(c)$

d. $f'(c)$

24 [Foundational] • Target: Cumulative Distribution Function $F(x)$

Given PDF $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, find the cumulative distribution function $F(x)$ for $0 \le x \le 2$.

a. $F(x) = \frac{x^3}{8}$

b. $F(x) = \frac{3x^3}{8}$

c. $F(x) = \frac{x^2}{4}$

d. $F(x) = \frac{x^3}{2}$

25 [Foundational] • Target: Cumulative Distribution Function $F(x)$

For the distribution in the previous question, find the median m.

a. $m = \sqrt[3]{4} \approx 1.587$

b. $m = 1.0$

c. $m = \sqrt{2} \approx 1.414$

d. $m = 1.5$

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26 [Foundational] • Target: Cumulative Distribution Function $F(x)$

Given $F(x) = \frac{x^2 - 1}{8}$ for $1 \le x \le 3$, find the PDF $f(x)$ on this interval.

a. $f(x) = \frac{x}{4}$

b. $f(x) = \frac{x}{8}$

c. $f(x) = \frac{x^2}{8}$

d. $f(x) = \frac{2x - 1}{8}$

27 [Foundational] • Target: Mean, Variance & Medians

Find the expected value $E(X)$ for the PDF $f(x) = 2x$ on $[0, 1]$.

a. $\frac{2}{3}$

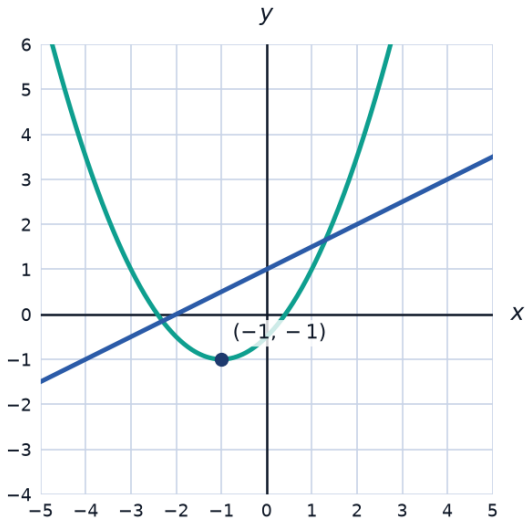
b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{1}{3}$

28 [Foundational] • Target: Mean, Variance & Medians

For the distribution $f(x) = 2x$ on $[0, 1]$, calculate the variance $\mathrm{Var}(X)$.



a. $\frac{1}{18}$

b. $\frac{1}{9}$

c. $\frac{1}{12}$

d. $\frac{2}{9}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Mean, Variance & Medians

Find the mode of the continuous distribution with PDF $f(x) = 12x^2(1-x)$ for $0 \le x \le 1$.

a. $\frac{2}{3}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{1}{3}$

30 [Intermediate] • Target: Piecewise Distributions & Examination Standards

A continuous random variable has PDF $f(x) = cx$ for $0 \le x < 2$ and $f(x) = c(4-x)$ for $2 \le x \le 4$. Find c.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{8}$

d. 1

31 [Intermediate] • Target: Piecewise Distributions & Examination Standards

For the symmetric triangular distribution in the previous question, what is $E(X)$?

a. 2

b. 1

c. 3

d. 2.5

32 [Intermediate] • Target: Piecewise Distributions & Examination Standards

Find the lower quartile Q_1 for the triangular distribution in the previous question ($c = 1/4$).

a. $\sqrt{2} \approx 1.414$

b. 1.0

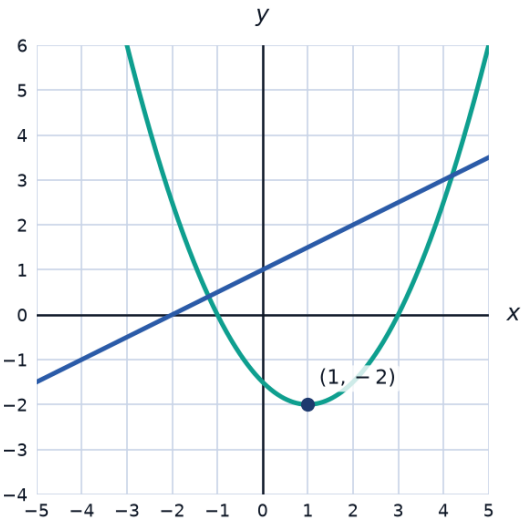
c. 1.5

d. $\sqrt{3} \approx 1.732$

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33 [Intermediate] • Target: Piecewise Distributions & Examination Standards

The continuous random variable X has probability density function given by $f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise}, \text{ where } k \text{ is a constant. Find the value of } k.$



a. $\frac{3}{16}$

b. $\frac{1}{4}$

c. $\frac{3}{8}$

d. $\frac{16}{3}$

34 [Intermediate] • Target: Verifying a valid probability density function

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function $f(x) = \{\frac{3}{8}x^2, , 0 \le x \le 2, ; 0, , \text{otherwise}.$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

35 [Intermediate] • Target: Finding an unknown constant in a pdf

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $\mathrm{P}(1 \le X \le 2)$.

a. $\frac{7}{8}$

b. $\frac{1}{8}$

c. $\frac{3}{4}$

d. $\frac{1}{2}$

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36 [Intermediate] • Target: Finding probabilities by integrating the pdf

The continuous random variable X has probability density function $f(x)$ defined by \[f(x) = \begin{cases} kx & 0 \le x 1 \mid X < 2.5) \).

a. $\frac{9}{11}$

b. $\frac{10}{11}$

c. $\frac{3}{4}$

d. $\frac{9}{10}$

37 [Intermediate] • Target: The cumulative distribution function

A continuous random variable X has PDF $f(x) = \frac{2}{10^2}x$ for $0 \le x \le 10$. Find the median of X in exact form.

a. $10\sqrt{2}$

b. $\frac{10}{2}$

c. $\frac{10\sqrt{2}}{2}$

d. $\frac{10\sqrt{2}}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 3.01 Advanced Trigonometry

A continuous random variable X has PDF $f(x) = \frac{2}{14^2}x$ for $0 \le x \le 14$. Find the median of X in exact form.

a. $14\sqrt{2}$

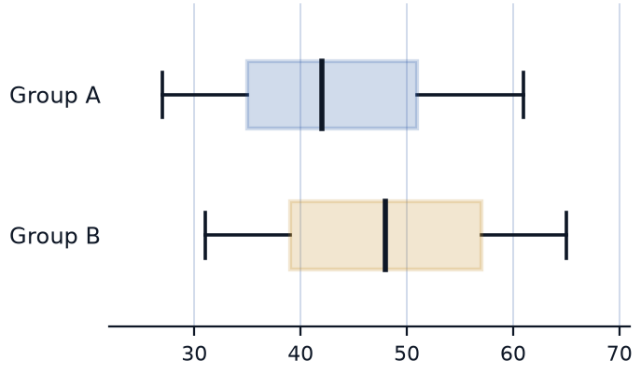
b. $\frac{14}{2}$

c. $\frac{14\sqrt{2}}{4}$

d. $\frac{14\sqrt{2}}{2}$

39 [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 4.01 Differentiation

A continuous random variable X has PDF $f(x) = \frac{2}{12^2}x$ for $0 \le x \le 12$. Find the median of X in exact form.



a. $\frac{12\sqrt{2}}{2}$

b. $\frac{12\sqrt{2}}{4}$

c. $\frac{12}{2}$

d. $12\sqrt{2}$

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40 [Challenging] • Target: Variance of a continuous random variable • Cross-Lesson: 5.01 Integration

A continuous random variable X has PDF $f(x) = \frac{2}{2^2}x$ for $0 \le x \le 2$. Find the median of X in exact form.

a. $\frac{2\sqrt{2}}{2}$

b. $\frac{2\sqrt{2}}{4}$

c. $\frac{2}{2}$

d. $2\sqrt{2}$

41 [Challenging] • Target: Median and quartiles • Cross-Lesson: 6.01 Sequences & Series

A continuous random variable X has PDF $f(x) = \frac{2}{6^2}x$ for $0 \le x \le 6$. Find the median of X in exact form.

a. $\frac{6\sqrt{2}}{4}$

b. $\frac{6}{2}$

c. $6\sqrt{2}$

d. $\frac{6\sqrt{2}}{2}$

42 [Challenging] • Target: A well-designed word problem: waiting time at a bus stop • Cross-Lesson: 7.01 Vectors

A continuous random variable X has PDF $f(x) = \frac{2}{8^2}x$ for $0 \le x \le 8$. Find the median of X in exact form.

a. $8\sqrt{2}$

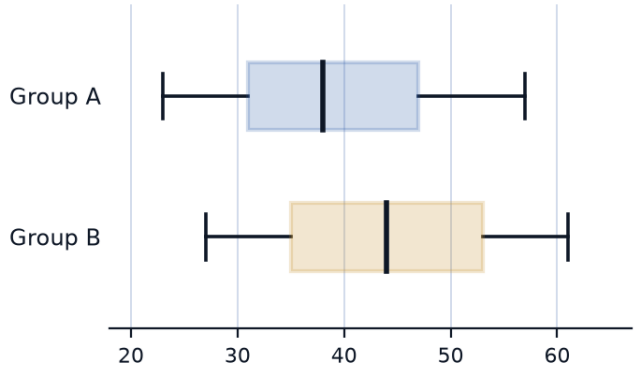
b. $\frac{8\sqrt{2}}{4}$

c. $\frac{8}{2}$

d. $\frac{8\sqrt{2}}{2}$

43 [Challenging] • Target: Probability Density Functions (PDF) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A continuous random variable X has PDF $f(x) = \frac{2}{4^2}x$ for $0 \le x \le 4$. Find the median of X in exact form.



a. $\frac{4\sqrt{2}}{4}$

b. $\frac{4}{2}$

c. $\frac{4\sqrt{2}}{2}$

d. $4\sqrt{2}$

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44 [Challenging] • Target: Cumulative Distribution Function $F(x)$ • Cross-Lesson: 9.01 Complex Numbers

A continuous random variable X has PDF $f(x) = \frac{2}{5^2}x^1$ for $0 \le x \le 5$. Find $\mathrm{Var}(X)$.

a. $\frac{10}{3}$

b. $\frac{25}{18}$

c. $\frac{25}{2}$

d. $\frac{43}{18}$

45 [Challenging] • Target: Mean, Variance & Medians • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A continuous random variable X has PDF $f(x) = \frac{3}{6^3}x^2$ for $0 \le x \le 6$. Find $\mathrm{Var}(X)$.

a. $\frac{108}{5}$

b. $\frac{9}{2}$

c. $\frac{27}{20}$

d. $\frac{47}{20}$

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